untitled

File Edit Format Run Options Window Help

```
def safe(cols, row, col):
    for r in range(row):
        c = cols[r]
        if c == col or abs(c - col) == abs(r - row):
            return False
    return True

def solve(n=8, find_all=True):
    solutions = []
    cols = [0] * n

    def place(row):
        if row == n:
            solutions.append(cols[:])
            return not find_all
        for col in range(n):
            if safe(cols, row, col):
                cols[row] = col
                if place(row + 1):
                    return True
        return False

    place(0)
    return solutions

def show(solution):
    n = len(solution)
    for r in range(n):
        print(" ".join("Q" if solution[r] == c else "." for c in range(n)))
    print()

if __name__ == "__main__":
    first = solve(8, find_all=False)[0]
    print("First solution (row -> column):", [c + 1 for c in first])
    show(first)
    print("Total number of solutions for 8 queens:", len(solve(8, find_all=True)))
```

Ln: 36 Col: 82

27°C Light rain

ENG IN 09:20 25-10-2026

IDLE Shell 3.11.1

File Edit Shell Debug Options Window Help

```
4 7 8

Step 3: move blank Left
1 _ 3
5 2 6
4 7 8

Step 4: move blank Down
1 2 3
5 _ 6
4 7 8

Step 5: move blank Left
1 2 3
_ 5 6
4 7 8

Step 6: move blank Down
1 2 3
4 5 6
_ 7 8

Step 7: move blank Right
1 2 3
4 5 6
7 _ 8

Step 8: move blank Right
1 2 3
4 5 6
7 8 _

Solved in 8 moves.
>>>
========== RESTART: C:/Users/ASUS/pyhon.py ==========
First solution (row -> column): [1, 5, 8, 6, 3, 7, 2, 4]
Q . . . . . .
. . . . Q . . .
. . . . . . . Q
. . . . . Q . .
. . Q . . . . .
. . . . . . Q .
. Q . . . . . .
. . . Q . . . .

Total number of solutions for 8 queens: 92
>>>
```

Ln: 64 Col: 0

27°C Light rain

ENG IN 09:20 25-10-2026